

Add functions

Use $f(x) = x^2$ and $g(x) = 2x - 1$.

$$(f + g)(x)$$

What formula represents the sum?

Subtract functions

Use $f(x) = x^2$ and $g(x) = 2x - 1$.

$$(f - g)(x)$$

What formula represents the difference?

Multiply functions

Use $f(x) = x^2$ and $g(x) = 2x - 1$.

$(fg)(x)$

What formula represents the product?

Divide functions

Use $f(x) = x^2$ and $g(x) = 2x - 1$.

$$\left(\frac{f}{g}\right)(x)$$

What formula and domain represent the quotient?

Shared input restrictions

Use $f(x) = \sqrt{x}$ and $g(x) = 1/(x - 2)$.

$$(f + g)(x)$$

What is the domain of the sum?

Composition

Composition uses the output of one function as the input of another.

$$(f \circ g)(x) = f(g(x))$$

Which function acts first?

Evaluate a composition

Use $f(x) = 3x + 1$ and $g(x) = x^2$.

$$f(g(-2))$$

What is the resulting output?

Reverse the order

Use $f(x) = 3x + 1$ and $g(x) = x^2$.

$$g(f(-2))$$

Does reversing the order change the result?

Compose formulas

Use $f(x) = x^2$ and $g(x) = x - 4$.

$$(f \circ g)(x)$$

What formula represents the composition?

Compare orders

Use $f(x) = x^2$ and $g(x) = x - 4$.

$$(g \circ f)(x)$$

How does this formula compare with $f(g(x))$?

Compose using data

The known values are $g(2) = 5$, $g(5) = 9$, $f(5) = 8$, and $f(9) = 3$.

$$f(g(2))$$

Which values are needed to evaluate this composition?

A composition with units

$A(q)$ is the area painted by q liters; $C(a)$ is the cost to prepare a square meters.

What does $C(A(3))$ represent?

A pricing composition

A \$50 item receives a \$10 discount and then 8% tax on the discounted price.

Which composition gives the final price?

Order in an application

A \$50 item receives 8% tax and then a \$10 discount from the taxed price.

How does this final price compare with discounting before tax?

The domain of a composition

An input must be allowed by the inner function, and its output must be allowed by the outer function.

Why can checking only the final simplified formula miss a restriction?

A radical composition

Use $f(x) = \sqrt{x}$ and $g(x) = x - 5$.

$$(f \circ g)(x)$$

What formula and real domain result?

A rational composition

Use $f(x) = 1/x$ and $g(x) = x^2 - 4$.

$$(f \circ g)(x)$$

What inputs are excluded?

An inner-function restriction

Use $f(x) = x^2$ and $g(x) = \sqrt{x - 1}$.

$$(f \circ g)(x)$$

After simplification, what domain must still be retained?

Both stages restrict inputs

Use $f(x) = \sqrt{x - 2}$ and $g(x) = 1/x$.

$$(f \circ g)(x)$$

What real input values make both stages defined?

Decompose a function

$$h(x) = \sqrt{3x + 1}$$

What inner and outer functions give h as a composition?

Decompose a rational function

$$h(x) = \frac{1}{(x - 2)^2}$$

What inner and outer functions give h as a composition?

Three stages

A rule subtracts 2, squares the result, and then adds 7.

What formula and composition represent the rule?